STPM Physics · Semester 2 · Chapter 16

Magnetic Fields

16.1 Concept of a Magnetic Field

Definitions & laws

- **Magnetic field** is a region in which a moving charge, a current-carrying conductor or a magnetic material experiences a magnetic force.
- **Magnetic field lines** are lines whose tangent at any point gives the direction of the magnetic field (flux density) at that point; outside a magnet they are directed from the north pole to the south pole.
- **Magnetic flux density, B,** is the force acting per unit length on a straight conductor carrying a current of 1 A that is placed perpendicular to the magnetic field.
- **One tesla** is the magnetic flux density that produces a force of 1 N on 1 m of straight conductor carrying a current of 1 A perpendicular to the field.
- Magnetic field lines always form closed loops (no magnetic monopoles); they never cross; the closer the lines, the stronger the field.
- A magnetic field is produced by a moving charge (a current) or by a permanent magnet; a stationary charge produces only an electric field.
- **Right-hand grip rule (straight wire):** grip the wire with the right hand, thumb pointing along the conventional current; the curled fingers give the direction of the field lines (concentric circles around the wire).

Formulae

Quantity / law	Formula	Symbols, units & conditions
Magnetic flux density	$B = \frac{F}{IL\sin\theta}$	B = magnetic flux density (T); F = force (N); I = current (A); L = length of conductor in the field (m); θ = angle between the conductor and B
Tesla	$1\ T = 1\ N\ A^{-1}\ m^{-1} = 1\ Wb\ m^{-2}$	T = tesla, the unit of magnetic flux density; Wb = weber
Magnetic field of a straight wire	$B = \frac{\mu_0 I}{2\pi r}$	I = current (A); r = perpendicular distance from the wire (m); $\mu_0 = 4\pi\times10^{-7}$ H m^{-1} (T m A^{-1})

How to answer (MPM style)

- Define B by quoting the force on a conductor: "force per unit current per unit length, conductor perpendicular to the field".
- When asked to sketch a field pattern: draw closed lines with arrows, closer together where the field is stronger, and label N and S poles or the current direction.
- For a wire, state "concentric circles in a plane perpendicular to the wire, getting further apart as r increases".

- State the direction rule used (right-hand grip) and the direction of the current in the same answer.

Section A traps

Trap: Field lines of a bar magnet "start at N and end at S" only outside the magnet; inside the magnet they continue from S to N, so the lines form closed loops.

Trap: A stationary charge has no magnetic field; a charge moving at constant velocity does.

Trap: Magnetic field lines are not the path of a charged particle and not the direction of the force; the force on a moving charge is perpendicular to B.

Tricks

Trick: For a straight wire $B \propto \frac{I}{r}$: doubling I and doubling r together leaves B unchanged; compare ratios instead of calculating.

Trick: Unit check: T = N A^{-1} m^{-1}; any option giving N A m^{-1} or N A m is wrong.

16.2 Force on a Moving Charge

Definitions & laws

- **Magnetic force on a moving charge:** a charge q moving with velocity v in a magnetic field B experiences a force of magnitude F = qvB sinθ, which is perpendicular to both v and B.
- The magnetic force is always perpendicular to the velocity, so it does no work and does not change the speed or kinetic energy of the particle; it changes only the direction of motion.
- When v is perpendicular to B, the magnetic force provides the centripetal force and the particle moves in a circle; when v is parallel (or antiparallel) to B, the force is zero and the motion is a straight line; at other angles the path is a helix.
- **Fleming's left-hand rule:** first finger = direction of B, second finger = direction of conventional current (direction of motion of a positive charge), thumb = direction of force. For a negative charge the second finger points opposite to its velocity.
- Positive and negative charges of the same velocity are deflected in opposite directions.
- **Velocity selector:** in perpendicular electric and magnetic fields, a particle passes undeflected when qE = qvB, so only particles of speed $v = \frac{E}{B}$ emerge.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Magnetic force on a moving charge	$F = qvB\sin\theta$	F = force (N); q = charge (C); v = speed (m s^{-1}); B = magnetic flux density (T); θ = angle between v and B
Force when v ⊥ B	$F = qvB$	Special case θ = 90°, i.e. v perpendicular to B
Circular motion in a magnetic field	$qvB = \frac{mv^2}{r}$	m = mass (kg); r = radius of the circular path (m)
Radius of circular path	$r = \frac{mv}{qB} = \frac{p}{qB}$	p = momentum (kg m s^{-1}); from $qvB = \frac{mv^2}{r}$
Period of circular motion	$T = \frac{2\pi m}{qB}$	T = period (s); independent of v and r
Frequency of circular motion	$f = \frac{qB}{2\pi m}$	f = frequency (Hz); independent of v and r
Velocity selector	$v = \frac{E}{B}$	E = electric field strength (V m^{-1}); B = flux density (T); v = speed (m s^{-1})
Energy gained from a p.d.	$qV = \frac{1}{2}mv^2$	V = accelerating potential difference (V); gain in kinetic energy equals work done by the p.d.
Speed after acceleration by a p.d.	$v = \sqrt{\frac{2qV}{m}}$	v = speed (m s^{-1}); q = charge (C); V = p.d. (V); m = mass (kg)
Constants for electron	$e = 1.60\times10^{-19}$ C; $m_e = 9.11\times10^{-31}$ kg	e = elementary charge (C); m_e = electron mass (kg)

How to answer (MPM style)

- Calculation: state $F = qvB\sin\theta$, substitute with SI units, give the magnitude with unit and the direction (Fleming's left-hand rule) where asked.
- For circular motion write "magnetic force = centripetal force, $qvB = \frac{mv^2}{r}$" before deriving $r = \frac{mv}{qB}$.
- State "the force is perpendicular to the velocity, hence no work is done and the speed is constant".
- In a ratio question, write $r \propto \frac{m}{q}$ (same v, B) or $r \propto v/\frac{q}{m}$ and substitute the ratios directly.

Section A traps

Trap: Using a negative charge in Fleming's rule without reversing the direction: the second finger points along the motion of a POSITIVE charge, so an electron's force is opposite to that of a proton moving the same way.

Trap: Alpha particle: charge +2e and mass ≈ $4m_p$, so $r_{\frac{\alpha}{r_p}} = \frac{4}{2} = 2$, not 4 or ½.

Trap: Velocity parallel to B gives zero force, not qvB; confusing sinθ with cosθ (angle taken to the normal).

Trap: Period of circular motion does not depend on speed: doubling v doubles r but T is unchanged.

Tricks

Trick: $r \propto \frac{mv}{qB}$: build a quick ratio table of m, v, q and B before computing anything.

Trick: Same p.d. accelerates different particles: $v \propto \sqrt{\frac{q}{m}}$; then $r \propto \sqrt{\frac{m}{q}}$ if B is the same.

16.3 Force on a Current-Carrying Conductor

Definitions & laws

- A straight conductor of length L carrying a current I in a magnetic field B experiences a force F = BIL sinθ, perpendicular to both the conductor and the field.
- **Fleming's left-hand rule:** first finger = field B, second finger = conventional current I, thumb = force F (motion); the three are mutually perpendicular.
- The force is a maximum (BIL) when the conductor is perpendicular to B and zero when the conductor is parallel to B.
- The force arises because the field exerts a force on each moving charge carrier; the sum over all the carriers in the length L gives the force on the conductor.
- A rectangular coil of N turns in a uniform field experiences no net force but a torque τ = NIAB sinφ, where φ is the angle between the normal to the coil and B (equivalently NIAB cosα where α is the angle between the plane of the coil and B).
- The torque is a maximum when the plane of the coil is parallel to B and zero when the plane is perpendicular to B; this is the principle of the moving-coil meter and the d.c. motor.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Force on a current-carrying wire	F = BIL sinθ	F = force (N); B = flux density (T); I = current (A); L = length of wire in the field (m); θ = angle between the wire and B
Torque on a coil	τ = NBIA sinφ	τ = torque (N m); N = number of turns; A = area of the coil (m^2); φ = angle between the normal to the coil and B

Quantity / law	Formula	Symbols, units & conditions
Wire balanced against its weight	BIL = mg	m = mass of wire (kg); g = acceleration of free fall (m s^{-2}); B perpendicular to the wire

How to answer (MPM style)

- State the formula, substitute in SI units (cm → m, g → kg) and quote the force in newtons with the direction from Fleming's left-hand rule.
- Direction: name the three directions used, e.g. "B into page, I to the right, so force is up the page".
- For a coil: state "net force = 0, but the two equal and opposite forces on opposite sides form a couple, giving a torque".
- For a balancing question equate magnetic force and weight, then solve for B or I.

Section A traps

Trap: Using L as the total length of wire instead of the length lying within the field.

Trap: Using the angle between the wire and B correctly: at 30°, sinθ = 0.5; do not use cos 30°.

Trap: Mixing up the plane and the normal of the coil: torque is maximum when the plane of the coil is PARALLEL to B (normal ⊥ B).

Trap: Reversing the current or field reverses the force; reversing both leaves it unchanged.

Tricks

Trick: Sketch axes x, y, z: if I and B are along two axes, the force lies along the third; the sign follows from I × B.

Trick: Limiting-case check: θ = 0° gives F = 0, so any option showing a non-zero force for a wire parallel to B is wrong.

16.4 Magnetic Fields Due to Currents

Definitions & laws

- **Right-hand grip rule (straight wire):** thumb along the conventional current, fingers curl in the direction of B.
- **Right-hand grip rule (coil/solenoid):** fingers curl in the direction of the current, thumb points along B inside the coil, i.e. towards the north-pole end.
- The field of a long straight wire consists of concentric circles; B is inversely proportional to the perpendicular distance r.

- The field inside a long solenoid (far from the ends) is uniform and parallel to the axis; the field outside is very small.
- Superposition: the resultant field at a point is the vector sum of the fields produced by each current; fields from different wires may cancel at a neutral point.
- Two coaxial coils or concentric coils with opposing currents give zero net field at the centre when their individual fields are equal in magnitude.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Field of a long straight wire	$B = \frac{\mu_0 I}{2\pi r}$	B = flux density (T); I = current (A); r = perpendicular distance from the wire (m)
Field at centre of a circular coil	$B = \frac{\mu_0 N I}{2r}$	N = number of turns; r = radius of the coil (m); I = current (A)
Field inside a long solenoid	$B = \mu_0 n I = \frac{\mu_0 N I}{l}$	n = number of turns per metre (m^{-1}); N = total number of turns; l = length of solenoid (m)
Field of a toroid	$B = \frac{\mu_0 N I}{2\pi r}$	r = mean radius of the ring (m); N = number of turns; I = current (A)
Permeability of free space	$\mu_0 = 4\pi \times 10^{-7}$ H m^{-1} = 1.26×10^{-6} H m^{-1}	μ_0 = permeability of free space (H m^{-1})

How to answer (MPM style)

- Quote the formula, substitute with SI units (cm → m) and give B in tesla to 3 s.f., with direction by the right-hand grip rule when asked.
- For a resultant field, find the magnitude and direction of each contribution separately, then add as vectors.
- For graphs of B against r (straight wire) sketch a curve of the form $\frac{1}{r}$: steep at small r, never reaching zero.
- For a solenoid always use $n = \frac{N}{l}$ (turns per metre), not N.

Section A traps

Trap: Using the total number of turns N in $B = \mu_0 n I$ instead of turns per metre.

Trap: Writing B for a straight wire $\propto \frac{1}{r^2}$ or $\propto r$; the correct relationship is $B \propto \frac{1}{r}$.

Trap: Rewinding the same length of wire into N turns: radius becomes $\frac{r_0}{N}$, so $B = \frac{\mu_0 N I}{2} \frac{r_0}{N}$ increases by N^2 (not N).

> **Trap:** Forgetting the 2π (wire) or the 2 (coil) in the denominator; the solenoid formula has neither.

Tricks

> **Trick:** Midway between two wires with currents in the same direction the fields cancel; for opposite directions they add.

> **Trick:** Same-length rewind: $B \propto N^2$; doubling the number of turns per metre of a solenoid doubles B.

16.5 Force Between Two Current-Carrying Conductors

Definitions & laws

- The field of one wire exerts a force on the current in the other; by Newton's third law the two forces are equal in magnitude and opposite in direction, whatever the sizes of the currents.
- Parallel wires carrying currents in the same direction attract each other; wires carrying currents in opposite directions repel each other.
- **The ampere** is that constant current which, if maintained in two straight parallel conductors of infinite length and negligible cross-section placed one metre apart in a vacuum, produces a force of 2×10^{-7} N per metre of length between them.
- Explanation: wire 1 produces a field $B_1 = \frac{\mu_0 I_1}{2\pi d}$ at wire 2; the force on wire 2 is $F = B_1 I_2 L$, directed by Fleming's left-hand rule towards wire 1 when the currents are parallel.
- **The tesla** is the magnetic flux density that gives a force of 1 N per metre on a conductor carrying 1 A perpendicular to the field.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Force per unit length between wires	$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$	$\frac{F}{L}$ = force per unit length (N m^{-1}); I_1 and I_2 = currents (A); d = separation (m)
Definition check (ampere)	$\frac{F}{L} = \frac{4\pi\times10^{-7}}{2\pi} = 2\times10^{-7}$ N m^{-1}	For $I_1 = I_2 = 1$ A and d = 1 m
Force on a length of wire	$F = \frac{\mu_0 I_1 I_2 l}{2\pi d}$	l = length of one wire (m); I_1 and I_2 = currents (A); d = separation (m)

How to answer (MPM style)

- Show the field of one wire at the position of the other, then F = BIL, then direction by Fleming's left-hand rule.
- State the nature: "attractive" for currents in the same direction, "repulsive" for opposite directions, with the magnitude per metre.

- In the definition of the ampere include all four conditions: constant current, infinite length and negligible cross-section, 1 m apart, in a vacuum; the force is 2×10^{-7} N per metre.
- Substitute with SI units and give $\frac{F}{L}$ in N m^{-1} to 3 s.f.; if a length l is given, multiply by l.

Section A traps

Trap: Same-direction currents REPEL: this is wrong; parallel currents attract (opposite currents repel).

Trap: Claiming the wire with the larger current feels the larger force; the two forces are equal and opposite.

Trap: Ratio questions: $\frac{F}{L} \propto \frac{I_1 I_2}{d}$, so doubling both currents and d gives $\frac{4}{2}$ = 2 times, not 4 or 1.

Trap: Using $\frac{\mu_0}{2\pi d}$ with d in cm; convert to metres first.

Tricks

Trick: Use the ratio method: write $F \propto \frac{I_1 I_2}{d}$ and substitute the multipliers.

Trick: Since $\frac{\mu_0}{2\pi} = 2\times10^{-7}$ T m A^{-1}, $\frac{F}{L} = 2\times10^{-7}\frac{I_1 I_2}{d}$ with d in metres; this avoids handling 4π.

16.6 Determination of the Ratio e/m

Definitions & laws

- **Specific charge** of the electron is the ratio of its charge to its mass, $\frac{e}{m}$ (C kg^{-1}).
- Thomson's method: electrons are accelerated and pass through crossed electric and magnetic fields; the fields are adjusted so that the beam is undeflected, so that eE = evB, giving $v = \frac{E}{B}$.
- With the electric field switched off, the magnetic force makes the electrons move in a circular arc: $evB = \frac{mv^2}{r}$, so $\frac{e}{m} = v/(Br)$.
- Alternative method: electrons accelerated through p.d. V gain kinetic energy $eV = \frac{1}{2}mv^2$; combined with $r = \frac{mv}{eB}$ this gives $\frac{e}{m} = 2V/(B^2r^2)$.
- The experiment determines the ratio $\frac{e}{m}$ only; the charge e must be found independently (Millikan's oil-drop experiment) to find m.
- The ratio $\frac{e}{m}$ for cathode rays is the same for all cathode materials, showing that electrons are a universal constituent of matter.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Balanced fields (velocity)	$eE = evB;\ v = \frac{E}{B}$	e = electron charge (C); E = electric field strength (V m^{-1}); B = flux density (T); v = speed (m s^{-1})
Magnetic deflection	$r = \frac{mv}{eB};\ \frac{e}{m} = \frac{v}{rB}$	r = radius of circular path (m); m = electron mass (kg)
Accelerating p.d.	$eV = \frac{1}{2}mv^2;\ v = \sqrt{\frac{2eV}{m}}$	V = accelerating p.d. (V)
Combined expression for $\frac{e}{m}$	$\frac{e}{m} = 2V/(B^2r^2)$	$\frac{e}{m}$ = specific charge of the electron (C kg^{-1})
Accepted value of $\frac{e}{m}$	$\frac{e}{m} = 1.76\times10^{11}$ C kg^{-1}	Accepted value of the specific charge (C kg^{-1})

How to answer (MPM style)

- Explain the first step: "balance electric and magnetic forces so that the beam is undeflected, hence $v = \frac{E}{B}$"; this gives the speed independently of $\frac{e}{m}$.
- Describe the second step: "remove the electric field; the beam moves in a circle of radius r; magnetic force provides centripetal force".
- For calculations write $\frac{e}{m} = 2V/(B^2r^2)$, substitute with SI units and give the answer to 3 s.f. in C kg^{-1} with the order of magnitude 10^{11}.
- For error analysis: $\frac{e}{m} \propto V/(B^2r^2)$, so a fractional error in B or r counts twice.

Section A traps

Trap: Forgetting that r and B are squared in $\frac{e}{m} = 2V/(B^2r^2)$; the squares also double fractional errors.

Trap: Giving the answer in the wrong power of ten: a result such as 10^{10} or 10^{12} C kg^{-1} signals an arithmetic slip; the accepted value is 1.76×10^{11} C kg^{-1}.

Trap: Claiming that $\frac{e}{m}$ experiments measure e or m alone; only the ratio is found.

Trap: Radius vs diameter: if the diameter of the circular path is measured, halve it before substituting.

Tricks

Trick: The $\frac{e}{m}$ value must be near 10^{11}: check the order of magnitude after calculating.

Trick: V is directly proportional to $\frac{e}{m}$: a reading of V too high by 10% gives $\frac{e}{m}$ too high by 10%.

16.7 Hall Effect

Definitions & laws

- **Hall effect** is the production of a potential difference (Hall voltage) across a current-carrying conductor or semiconductor placed in a magnetic field, the potential difference being perpendicular to both the current and the field.
- Explanation: the magnetic force deflects the moving charge carriers to one side of the slab; the accumulated charge sets up a transverse electric field E_H.
- The Hall voltage reaches a steady value when the electric force on a carrier balances the magnetic force: $eE_H = evB$.
- The sign of the Hall voltage identifies the sign of the charge carriers: with the same current direction and field direction, electrons and holes are deflected to the same side but give opposite polarity across the slab (use Fleming's left-hand rule with conventional current).
- In a semiconductor the carrier density n is much lower than in a metal, so V_H is much larger.
- Application: a Hall probe measures B because $V_H \propto B$ at constant current; it is made from a thin semiconductor slab for a large V_H.

Formulae

Quantity / law	Formula	Symbols, units & conditions
Hall voltage	$V_H = \frac{BI}{nte}$	V_H = Hall voltage (V); B = flux density (T); I = current (A); n = number density of carriers (m^{-3}); t = thickness of the slab in the direction of B (m); e = 1.60×10^{-19} C
Balance of forces	$\frac{eV_H}{d} = evB$	d = width of the slab between the Hall faces (m); v = drift velocity ($m\ s^{-1}$)
Current in terms of drift velocity	$I = nevtd$	Substituting into the force balance gives $V_H = \frac{BI}{nte}$
Drift velocity	$v = \frac{I}{nAe}$	A = cross-sectional area (m^2); n = number density of carriers (m^{-3}); e = 1.60×10^{-19} C

How to answer (MPM style)

- Explain by steps: carriers move, magnetic force deflects them to one face, charge accumulates, transverse E field builds up until $eE_H = evB$.

- State the polarity: for electrons deflected by the field, the face where they accumulate is negative; for holes the opposite face becomes positive; mention the direction rule used.
- For calculations write $V_H = \frac{BI}{nte}$, substitute with SI units (mm → m) and give V_H in volts (μV or mV) to 3 s.f.
- Explain why a semiconductor is used: low n gives large V_H (and a thin slab, small t, also raises V_H).

Section A traps

Trap: t is the thickness in the direction of B, not the width of the strip; using the wrong dimension gives the wrong V_H.

Trap: Taking $V_H \propto n$: V_H is inversely proportional to n, so a low density of carriers gives a LARGER Hall voltage.

Trap: Forgetting to convert mm to m (0.10 mm = 1.0×10^{-4} m); metals give V_H of order μV.

Trap: The Hall voltage does not depend on the length of the slab or the sign of the carriers in magnitude; only the polarity depends on the sign.

Tricks

Trick: $V_H \propto \frac{BI}{nt}$: doubling I doubles V_H; doubling t halves V_H.

Trick: Estimate order of magnitude: for metals $n \approx 10^{29}$ m^{-3} gives $V_H \approx$ μV; semiconductors ($n \approx 10^{20}$ m^{-3}) give mV.